BLUESTOCK FINTECH

Capstone Project

Mutual Fund Analytics Platform

End-to-End Data Engineering, ETL Pipeline & Interactive Dashboard

Company	Bluestock Fintech Pvt. Ltd.
Domain	Mutual Fund / Fintech
Project Type	Individual Capstone (1 Week)
Duration	7 Working Days | ~50-55 Hours
Data Source	AMFI India (Public), mfapi.in, NSE/BSE Public Data
Technologies	Python, SQL, Power BI / Tableau, Pandas, Matplotlib
Submission	PDF Report + Dashboard + GitHub Repository
Prepared By	Intern / Data Analyst — Bluestock Fintech
Date	June 2026

40 Schemes	10 Datasets	87K+ Rows	4.5 Yrs
Real Fund Schemes	Provided	Transaction Data	NAV History

Table of Contents

1. Project Overview & Objective

Bluestock Fintech is a financial technology company focused on democratising investment analytics for retail and institutional investors in India. This capstone project tasks you with building a **full-stack Mutual Fund Analytics Platform** that ingests publicly available data from AMFI India, transforms it through a robust ETL pipeline, stores it in a relational database, and presents insights via an interactive dashboard.

1.1 Business Context

The Indian mutual fund industry has grown at a remarkable pace. As of December 2025, the industry manages over **Rs. 81 lakh crore** in AUM across 1,908 schemes and **26.12 crore investor folios**. Monthly SIP inflows crossed an all-time high of **Rs. 31,002 crore** in December 2025, reflecting India's deepening equity culture. Bluestock Fintech wants a data platform that:

- Tracks NAV movements of 40+ key mutual fund schemes from top AMCs
- Monitors AUM growth trends for the 10 largest fund houses over 4+ years
- Analyses investor behaviour patterns across geographies and demographics
- Computes risk-adjusted return metrics (Sharpe, Sortino, Alpha, Beta)
- Benchmarks fund performance against Nifty 50, Nifty 100, BSE SmallCap indices
- Provides an executive-level interactive dashboard for fund selection

1.2 Project Objectives

#	Objective	Outcome
O1	Build an ETL pipeline from raw AMFI data	Automated Python script
O2	Design a normalised SQL schema for MF data	5-table star schema
O3	Perform comprehensive EDA on NAV & AUM data	EDA notebook with charts
O4	Compute performance & risk metrics per scheme	Metrics dashboard
O5	Build an interactive BI dashboard	Power BI / Tableau file
O6	Analyse investor transaction patterns	Demographic insights
O7	Compare fund returns vs benchmark indices	Alpha / tracking error report
O8	Document and present the entire project	PDF report + slides

2. Problem Statement

Despite the massive growth of India's mutual fund industry, individual investors and financial advisors often struggle to make data-driven fund selection decisions due to fragmented data, lack of unified analytics platforms, and complex financial metrics. This project solves real business problems:

P1: Data Fragmentation

Problem	NAV data, AUM data, SIP flow data and portfolio holdings are available on different sections of the AMFI website in different formats (TXT, PDF, HTML tables). No single unified database exists.
Solution	**Build an ETL pipeline that consolidates all data into a single SQLite/PostgreSQL database.**

P2: Performance Comparison Gap

Problem	Investors cannot easily compare multiple funds across different AMCs on a risk-adjusted basis. Raw NAV data requires significant transformation to compute Sharpe ratio, Alpha, Beta, etc.
Solution	**Compute and visualise all key risk-return metrics in a single dashboard.**

P3: No Benchmark Tracking

Problem	Most retail investors do not know whether their fund is outperforming its benchmark index. Tracking error and information ratio require both NAV and benchmark index data.
Solution	**Join NAV data with benchmark index prices and compute rolling alpha and tracking error.**

P4: Investor Behaviour Blind Spot

Problem	AMCs and distributors have limited visibility into how demographic and geographic factors influence SIP amounts, fund preferences and redemption patterns.
Solution	**Analyse investor transaction data to generate demographic segmentation and behavioural insights.**

P5: Slow Reporting

Problem	Monthly MF reports are static PDFs that take days to prepare. Stakeholders need live, self-service dashboards with drill-down capability.
Solution	**Build a Power BI / Tableau dashboard refreshable from the ETL pipeline output.**

3. Data Sources & Datasets

All datasets used in this project are derived from **publicly available, freely accessible sources**. No proprietary or confidential data is used. The primary sources are:

Source	URL / API	Data Type	Update Freq.
AMFI India	www.amfiindia.com	NAV, AUM, Folio, SIP	Daily / Monthly
mfapi.in	api.mfapi.in/mf/{code}	Historical NAV (JSON)	Daily
mfdata.in	mfdata.in/api/v1/schemes/	NAV + Expense Ratio	Daily
NSE India	nseindia.com/reports	Benchmark Index Prices	Daily
BSE India	bseindia.com	BSE SmallCap Index	Daily
AMFI Monthly Notes	amfiindia.com/research	Industry SIP & Flow Data	Monthly

3.1 Dataset Inventory

01_fund_master.csv	**40 rows**
Master list of 40 real mutual fund schemes (AMFI codes, fund house, category, expense ratio, risk grade, fund manager)	
02_nav_history.csv	**~46,000 rows**
Daily NAV for all 40 schemes from Jan 2022 to May 2026, anchored to real AMFI NAV values from mfapi.in	
03_aum_by_fund_house.csv	**~90 rows**
Quarterly AUM (in Rs. crore) for 10 fund houses from 2022 to 2025, sourced from real AMFI quarterly reports	
04_monthly_sip_inflows.csv	**48 rows**
Month-wise SIP inflow (Rs. crore), active SIP accounts, new registrations and SIP AUM — real AMFI Monthly Note data	
05_category_inflows.csv	**~144 rows**
Net inflows by fund category (Large Cap, Mid Cap, Small Cap, ELSS, Liquid, etc.) for FY 2024-25	
06_industry_folio_count.csv	**21 rows**
Total mutual fund folios (in crore) broken by Equity, Debt, Hybrid — real AMFI published milestones	
07_scheme_performance.csv	**40 rows**

1yr/3yr/5yr returns, Sharpe, Sortino, Alpha, Beta, Max Drawdown, Std Dev — computed from NAV history

08_investor_transactions.csv	**~32,000 rows**
Simulated SIP + Lumpsum + Redemption transactions for 5,000 investors across Indian states with demographics	

09_portfolio_holdings.csv	**~320 rows**
Top equity holdings (stock, weight %, sector) for equity mutual funds as of Dec 2025	

10_benchmark_indices.csv	**~8,000 rows**
Daily closing values for Nifty 50, Nifty 100, Nifty Midcap 150, BSE SmallCap, CRISIL Liquid & Gilt indices	

3.2 Key Real-World Data Points Embedded

Metric	Value	Source
SBI MF AUM Dec 2025	Rs. 12.50 lakh crore (largest AMC in India)	AMFI Quarterly
ICICI Pru MF AUM Dec 2025	Rs. 10.74 lakh crore	AMFI Quarterly
HDFC MF AUM Dec 2025	Rs. 9.30 lakh crore	AMFI Quarterly
SIP Inflow Dec 2025	Rs. 31,002 crore (all-time high at time of project)	AMFI Monthly Note
Active SIP Accounts Dec 2025	9.35 crore accounts	AMFI Monthly Note
Total MF Folios Dec 2025	26.12 crore	AMFI / Business Standard
Industry AUM Dec 2025	Rs. 81 lakh crore (Oct 2025 peak: Rs. 79.9 lakh crore)	AMFI
NAV anchor: HDFC Top 100	Rs. 892.45 (Oct 2024) — from mfapi.in code 125497	mfapi.in

4. System Architecture & ETL Pipeline

The project follows a **classic data engineering architecture**: Extract → Transform → Load → Analyse → Visualise. This mirrors real-world fintech data pipelines used at companies like Zerodha, Groww, and Paytm Money.

LAYER 1: DATA SOURCES (Extract)

- AMFI Daily NAV TXT (amfiindia.com/spages/NAVAll.txt)
- AMFI Historical NAV API (portal.amfiindia.com/DownloadNAVHistoryReport_Po.aspx)
- mfapi.in REST API (GET /mf/{scheme_code} — JSON, no auth required)
- AMFI Monthly Notes PDF (amfiindia.com/research-information/amfi-monthly)
- NSE/BSE Bhavcopy CSV (nseindia.com/reports)
- Pre-packaged CSV datasets provided with this project

LAYER 2: DATA PROCESSING (Transform)

- Python (Pandas) — parse, clean, reshape, merge datasets
- Handle missing NAV values (weekends/holidays → forward-fill)
- Compute derived fields: daily returns, rolling averages, CAGR
- Normalise fund names (AMFI naming inconsistencies)
- Validate AMFI codes against master list
- Enrich transactions with fund metadata (join on amfi_code)

LAYER 3: DATA STORAGE (Load)

- SQLite (development) or PostgreSQL (production)
- 5-table star schema: dim_fund, dim_date, fact_nav, fact_transactions, fact_performance
- Indexed on amfi_code, date for fast query performance
- CSV flat-file backup for Tableau / Power BI direct import

LAYER 4: ANALYTICS (Analyse)

- Jupyter Notebooks for EDA (Matplotlib, Seaborn, Plotly)
- Risk metrics: Sharpe, Sortino, Alpha, Beta, Max Drawdown, VaR
- Benchmark comparison: rolling 3-month alpha vs Nifty 100
- Cohort analysis: investor tenure vs average SIP amount
- Geographic heatmap: T30 vs B30 city AUM contribution

LAYER 5: VISUALISATION (Dashboard)

- Power BI / Tableau Desktop dashboard
- 4 report pages: Fund Overview, Performance, Investor, Industry Trends
- Slicers: Fund House, Category, Date Range, State
- KPI cards: AUM, SIP Inflow, Folio Count, Top Performer
- Export to PDF for stakeholder distribution

4.1 Database Schema (Star Schema)

Table	Type	Key Columns	Rows (approx.)
`dim_fund`	Dimension	amfi_code (PK), fund_house, category, expense_ratio	40

`dim_date`	Dimension	date_id (PK), date, year, month, quarter, is_weekday	1,500
`fact_nav`	Fact	amfi_code (FK), date (FK), nav, daily_return_pct	46,000
`fact_transactions`	Fact	tx_id (PK), investor_id, amfi_code (FK), date, amount, type	32,000+
`fact_performance`	Fact	amfi_code (FK), as_of_date, return_1yr, sharpe, alpha, max_dd	40
`fact_portfolio`	Fact	amfi_code (FK), stock_symbol, weight_pct, sector, date	320
`fact_aum`	Fact	fund_house, date, aum_crore, num_schemes	90
`fact_sip_industry`	Fact	month, sip_inflow_crore, sip_accounts_crore	48

5. 7-Day Task Breakdown

This project is designed to be completed **individually in 7 working days (~50–55 hours)**. Each day has a clear focus area, specific tasks, required tools, and concrete deliverables. Days 1-2 are foundational (data engineering), Days 3-4 are analytical, Days 5-6 are visualisation/advanced analytics, and Day 7 is documentation and presentation.

DAY 1: Project Setup + Data Ingestion (ETL)

Week Day 1 | 6–8 hours

Objectives

- Set up the project environment (Python, VS Code / Jupyter, Git)
- Understand the AMFI data ecosystem and public APIs
- Write Python scripts to ingest all 10 provided CSV datasets
- Fetch live NAV data from mfapi.in API for 5 selected schemes
- Create a Git repository and commit the initial project structure

#	Task	Tools / Tech	Output / Deliverable	Time
1	Create project folder structure: project/ ■■■ data/raw/ ■■■ data/processed/ ■■■ notebooks/ ■■■ sql/ ■■■ dashboard/ ■■■ reports/	`VS Code, Git`	*Folder structure committed to GitHub*	30m
2	Install dependencies: pip install pandas numpy matplotlib seaborn plotly sqlalchemy sqlite3 requests jupyter	`pip, requirements.txt`	*requirements.txt file*	20m
3	Load all 10 CSV datasets using pandas, print shape/dtypes/head for each	`Python, Pandas`	*data_ingestion.py script*	60m
4	Fetch live NAV from mfapi.in: GET https://api.mfapi.in/mf/125497 (HDFC Top 100) Parse JSON response and save as raw CSV	`requests, json, Pandas`	*live_nav_fetch.py + raw CSV*	45m
5	Fetch NAV for 5 schemes: SBI Bluechip (119551), ICICI Bluechip (120503), Nippon Large Cap (118632), Axis Bluechip (119092), Kotak Bluechip (120841)	`mfapi.in REST API`	*5 raw NAV CSV files*	60m
6	Understand fund master: print unique fund houses, categories, sub-categories, risk grades	`Pandas, Jupyter`	*EDA cell outputs in notebook*	30m
7	Validate AMFI codes — check all codes in fund_master exist in nav_history	`Pandas merge/join`	*Data quality report (text)*	20m

#	Task	Tools / Tech	Output / Deliverable	Time
8	Git commit with message: 'Day 1: Data ingestion complete'	`Git, GitHub`	*Remote repository updated*	10m

Day 1 Deliverables: Project repo, data_ingestion.py, live_nav_fetch.py, 15 raw CSV files in data/raw/

DAY 2: Data Cleaning + SQL Database Design

Week Day 2 | 7–8 hours

Objectives

- Clean and validate all 10 datasets (handle nulls, duplicates, type errors)
- Design and implement a 5+ table SQLite database schema
- Load all cleaned data into the database
- Write and test 10 SQL queries for basic analytics

#	Task	Tools / Tech	Output / Deliverable	Time
1	Clean nav_history.csv: - Parse dates to datetime - Sort by amfi_code + date - Forward-fill missing NAV (holidays) - Remove duplicates - Validate NAV > 0	`Pandas, datetime`	*clean_nav.csv (46K rows)*	75m
2	Clean investor_transactions.csv: - Standardise transaction_type (SIP/Lumpsum/Redemption) - Validate amount > 0 - Check KYC status values - Fix date formats	`Pandas, re`	*clean_transactions.csv*	60m
3	Clean scheme_performance.csv: - Validate return values are numeric - Flag negative Sharpe ratios - Check expense_ratio range (0.1% – 2.5%)	`Pandas, numpy`	*clean_performance.csv*	30m
4	Design SQLite schema: CREATE TABLE dim_fund (amfi_code TEXT PK, fund_house, scheme_name, category, ...) CREATE TABLE fact_nav (amfi_code TEXT FK, nav_date DATE, nav REAL, daily_return REAL) CREATE TABLE fact_transactions (...) CREATE TABLE fact_performance (...)	`SQLite, SQL DDL`	*schema.sql file*	60m
5	Load all cleaned datasets into SQLite: engine = create_engine('sqlite:///bluestock_mf.db') df.to_sql('fact_nav', engine, ...)	`SQLAlchemy, SQLite`	*bluestock_mf.db (database file)*	45m

#	Task	Tools / Tech	Output / Deliverable	Time
6	Write 10 SQL queries: 1. Top 5 funds by AUM 2. Average NAV per month 3. SIP inflow YoY growth 4. Transactions by state 5. Funds with expense_ratio < 1% 6-10: (see notebook)	`SQL, SQLite CLI`	*queries.sql with results*	60m
7	Create data dictionary documenting all columns, types, and sources	`Markdown/Excel`	*data_dictionary.md*	30m
8	Git commit: 'Day 2: Cleaned data + SQLite DB loaded'	`Git`	*GitHub updated*	10m

Day 2 Deliverables: 10 clean CSVs, bluestock_mf.db, schema.sql, queries.sql, data_dictionary.md

DAY 3: Exploratory Data Analysis (EDA)

Week Day 3 | 7–8 hours

Objectives

- Perform deep EDA on NAV, AUM, SIP and investor data
- Create 15+ publication-quality charts using Matplotlib/Seaborn/Plotly
- Identify key trends, anomalies and insights
- Document findings in a structured Jupyter Notebook

#	Task	Tools / Tech	Output / Deliverable	Time
1	NAV trend analysis: Plot daily NAV for all 40 schemes 2022–2026. Highlight COVID recovery, 2023 rally, 2024 corrections.	`Matplotlib, Plotly`	*Chart: NAV Trend Lines*	60m
2	AUM growth bar chart: Grouped bar chart — AUM by fund house for each year 2022–2025. Highlight SBI's dominance at Rs.12.5L Cr.	`Seaborn barplot`	*Chart: AUM Growth by AMC*	45m
3	SIP inflow time-series: Monthly SIP inflow Jan 2022 to Dec 2025. Mark the Rs.31,002 Cr milestone (Dec 2025).	`Plotly line chart`	*Chart: SIP Inflow Trend*	30m
4	Category-wise inflow heatmap: Months on X-axis, categories on Y-axis, net inflow as colour intensity.	`Seaborn heatmap`	*Chart: Category Heatmap*	45m
5	Investor demographics: Age group distribution pie chart. SIP amount box plot by age group.	`Matplotlib, Seaborn`	*2 Charts: Demographics*	45m
6	Geographic distribution: Horizontal bar chart — SIP amount by state. T30 vs B30 pie chart.	`Matplotlib, geopandas (optional)`	*Chart: Geo Distribution*	60m
7	Folio count growth: Line chart Jan 2022 to Dec 2025 showing growth from 13.26 to 26.12 crore.	`Plotly`	*Chart: Folio Count Growth*	30m
8	Correlation matrix: Compute pairwise correlation of NAV returns across 10 selected funds.	`Seaborn heatmap, numpy`	*Chart: Correlation Matrix*	45m
9	Top holdings sector distribution: Pie/donut chart of sector weights across all equity fund portfolios.	`Matplotlib donut`	*Chart: Sector Allocation*	30m
10	Summarise 10 key EDA findings as bullet points in notebook markdown cells	`Jupyter Markdown`	*EDA_Findings.ipynb*	30m

Day 3 Deliverables: EDA_Analysis.ipynb with 15+ charts, EDA_Findings summary, exported PNG charts

DAY 4: Fund Performance Analytics

Week Day 4 | 7–8 hours

Objectives

- Compute all key performance and risk metrics from NAV history
- Build a fund ranking/scoring model
- Compare fund returns against benchmark indices
- Identify best and worst performing funds per category

#	Task	Tools / Tech	Output / Deliverable	Time
1	Compute daily returns for all funds: daily_return = nav_t / nav_t-1 - 1 Annualised return = (1 + daily_return).prod()^(252/n) - 1	`Pandas, numpy`	*returns_computed.csv*	45m
2	Calculate CAGR for 1yr, 3yr, 5yr periods: CAGR = (NAV_end / NAV_start) ^ (1/n) - 1 for SBI Bluechip, HDFC Top 100, etc.	`Pandas groupby, numpy`	*cagr_report.csv*	60m
3	Compute Sharpe Ratio: Sharpe = (Rp - Rf) / Std(Rp) Use Rf = 6.5% (RBI repo rate proxy) Annualise with sqrt(252)	`numpy, Pandas`	*sharpe_values.csv*	45m
4	Compute Sortino Ratio: Sortino = (Rp - Rf) / Downside_Std where Downside_Std uses only negative return days	`numpy`	*sortino_values.csv*	30m
5	Compute Alpha & Beta vs benchmark: Regress fund returns on Nifty 100 returns (OLS) Alpha = intercept * 252, Beta = slope Use scipy.stats.linregress	`scipy, numpy`	*alpha_beta.csv*	60m
6	Compute Maximum Drawdown: max_dd = min(NAV / running_max - 1) Highlight worst drawdown period for each fund	`Pandas cummax, numpy`	*max_drawdown.csv*	45m
7	Build Fund Scorecard (composite score 0-100): Score = 30%×(3yr return rank) + 25%×(Sharpe rank) + 20%×(Alpha rank) + 15%×(Expense ratio rank, inverse) + 10%×(Max DD rank, inverse)	`Pandas rank, weighted sum`	*fund_scorecard.csv*	60m
8	Benchmark comparison chart: Plot top 5 funds vs Nifty 50 and Nifty 100 over 3 years. Compute tracking error.	`Matplotlib, numpy std`	*benchmark_chart.png*	45m

Day 4 Deliverables: Performance_Analytics.ipynb, fund_scorecard.csv, alpha_beta.csv, benchmark comparison chart

DAY 5: Dashboard Development (Power BI / Tableau)

Week Day 5 | 7–8 hours

Objectives

- Connect Power BI / Tableau to the SQLite database and cleaned CSVs
- Build 4 interactive dashboard pages
- Add slicers, drill-through, and tooltips
- Publish / export the dashboard

#	Task	Tools / Tech	Output / Deliverable	Time
1	Connect Power BI to bluestock_mf.db (SQLite ODBC) OR import CSVs directly. Verify all 8 tables load correctly.	`Power BI / Tableau`	*Data model loaded*	45m
2	Build PAGE 1 — Industry Overview: KPI Cards: Total AUM (Rs.81L Cr), SIP Inflows (Rs.31K Cr), Folios (26.12 Cr), # Schemes (1908) Line chart: Industry AUM Jan2022-Dec2025 Bar chart: AUM by fund house (top 10)	`Power BI`	*Dashboard Page 1*	75m
3	Build PAGE 2 — Fund Performance: Scatter plot: Return (X) vs Risk/StdDev (Y), bubble=AUM Table: Sortable fund scorecard Line chart: NAV of selected fund vs benchmark Slicer: Fund House, Category, Plan	`Power BI`	*Dashboard Page 2*	75m
4	Build PAGE 3 — Investor Analytics: Map/Bar: Transaction amount by state Donut: SIP vs Lumpsum vs Redemption split Bar: Age group vs avg SIP amount Line: Monthly transaction volume Slicer: State, Age Group, City Tier	`Power BI`	*Dashboard Page 3*	75m
5	Build PAGE 4 — SIP & Market Trends: Dual-axis: SIP Inflow (bar) + Nifty 50 (line) 2022-2025 Heat map: Category inflows by month Bar: Top 5 categories by net inflow FY25 KPI: SIP accounts growth YoY	`Power BI`	*Dashboard Page 4*	60m
6	Add tooltips, drill-through from fund table to NAV chart. Add Bluestock logo and colour theme.	`Power BI formatting`	*Branded dashboard*	30m
7	Export dashboard to PDF. Export each page as PNG for the final report.	`Power BI Export`	*Dashboard.pdf + 4 PNGs*	20m

Day 5 Deliverables: bluestock_mf_dashboard.pbix (Power BI file), Dashboard.pdf, 4 page screenshots

DAY 6: Advanced Analytics + Risk Metrics

Week Day 6 | 6–7 hours

Objectives

- Implement Value at Risk (VaR) and Conditional VaR
- Perform investor cohort analysis
- Build a simple fund recommendation model
- Analyse sector concentration risk in portfolios

#	Task	Tools / Tech	Output / Deliverable	Time
1	Compute Historical VaR (95%) for each fund: VaR = 5th percentile of daily return distribution CVaR = mean of returns below VaR threshold	`numpy percentile, Pandas`	*var_cvar_report.csv*	60m
2	Compute Rolling 90-day Sharpe Ratio for 5 funds: rolling_sharpe = returns.rolling(90).mean() / returns.rolling(90).std() * sqrt(252) Plot over time	`Pandas rolling, Matplotlib`	*rolling_sharpe_chart.png*	45m
3	Investor cohort analysis: Group investors by first transaction year (2024/2025) Compute average SIP amount, total invested, fund preference by cohort	`Pandas groupby, pivot`	*cohort_analysis.csv*	60m
4	SIP continuation analysis: For each investor with 6+ SIP transactions, compute average gap between transactions. Flag investors with gaps > 35 days as 'at-risk'	`Pandas date diff, groupby`	*sip_continuity.csv*	45m
5	Simple fund recommendation logic: Input: investor risk appetite (Low/Moderate/High) Output: Top 3 funds by Sharpe ratio within matching risk_grade Print recommendation table	`Python functions, Pandas filter`	*recommender.py*	45m
6	Sector concentration analysis: For each equity fund, compute Herfindahl-Hirschman Index (HHI) of sector weights. HHI = sum(weight_i^2). High HHI = concentrated portfolio.	`numpy, Pandas groupby`	*sector_hhi.csv + chart*	45m
7	Write advanced analytics summary with 5 key insights (e.g. which funds have highest VaR, which cohorts invest most, etc.)	`Jupyter Markdown`	*Advanced_Analytics.ipynb*	30m

Day 6 Deliverables: Advanced_Analytics.ipynb, var_cvar_report.csv, rolling_sharpe_chart.png, recommender.py

DAY 7: Final Report + Presentation + Deployment

Week Day 7 | 6–7 hours

Objectives

- Write the comprehensive final project report (PDF)
- Create a 12-slide presentation deck
- Clean and document all code (README, docstrings)
- Push final code to GitHub with proper README
- Optional: Deploy dashboard to Power BI Service / Tableau Public

#	Task	Tools / Tech	Output / Deliverable	Time
1	Write Final Report (this document serves as template): Sections: Executive Summary, Data, ETL, EDA Findings, Performance Analysis, Dashboard Screenshots, Recommendations, Limitations	`Word / LaTeX / Python ReportLab`	*Final_Report.pdf (15–20 pages)*	120 m
2	Create 12-slide presentation deck: Slide 1: Title Slide 2: Problem & Objective Slide 3: Data Sources Slide 4: Architecture Slide 5-6: EDA Highlights Slide 7-8: Performance Metrics Slide 9-10: Dashboard Screenshots Slide 11: Key Findings Slide 12: Thank You	`PowerPoint / Google Slides`	*Bluestock_MF_Presentation.pptx*	90m
3	Clean all Python scripts: Add docstrings, remove debug prints Add README.md to each folder Create master run script: python run_pipeline.py	`VS Code, Python`	*Clean codebase*	45m
4	Write root README.md: Project overview, setup instructions, file descriptions, how to run ETL, how to open dashboard	`Markdown, Git`	*README.md*	30m
5	Final GitHub push: git add . git commit -m 'Final: Complete Bluestock MF Capstone' git push origin main Tag release: git tag v1.0	`Git, GitHub`	*Public GitHub repo*	20m
6	Optional: Publish Power BI dashboard to Power BI Service (free account). OR publish to Tableau Public: File → Publish to Tableau Public	`Power BI Service / Tableau Public`	*Public dashboard URL*	30m
7	Self-review checklist: ✓ All 8 objectives met? ✓ All 7 deliverables submitted? ✓ Code runs without errors? ✓ Dashboard loads correctly? ✓ Report is professional?	`Checklist`	*Submission-ready package*	15m

Day 7 Deliverables: Final_Report.pdf, Presentation.pptx, clean GitHub repo with README, dashboard published

6. Technical Stack Details

Category	Tool / Library	Version	Purpose
Language	`Python`	3.10+	All data processing, ETL, analytics
Data Manipulation	`Pandas`	2.0+	DataFrames, cleaning, aggregation
Numerical	`NumPy`	1.24+	Risk metrics, statistical functions
Visualisation	`Matplotlib`	3.7+	Static charts for reports
Visualisation	`Seaborn`	0.12+	Heatmaps, distributions
Visualisation	`Plotly`	5.x	Interactive charts in Jupyter
Database	`SQLite3`	Built-in	Local relational database
ORM	`SQLAlchemy`	2.0	Python-to-SQLite interface
Statistics	`SciPy`	1.10+	OLS regression for Beta/Alpha
Notebooks	`Jupyter Lab`	4.x	EDA and analytics notebooks
Dashboard	`Power BI Desktop`	Latest	Interactive BI dashboard
Dashboard (alt)	`Tableau Desktop`	2024.x	Alternative to Power BI
Version Control	`Git + GitHub`	Latest	Code versioning, submission
API	`mfapi.in`	v1	Live NAV data (no auth required)
API (alt)	`mfdata.in`	v1	NAV + expense ratio data
HTTP Client	`Requests`	2.30+	API calls from Python
IDE	`VS Code / PyCharm`	Latest	Code development
Reporting	`ReportLab / Word`	4.x	Final PDF report

6.1 Folder Structure

```
bluestock_mf_capstone/ ■■■ data/ ■ ■■■ raw/ ← Original downloaded files ■ ■■■ processed/
← Cleaned, merged CSVs ■ ■■■ db/ ← bluestock_mf.db (SQLite) ■■■ notebooks/ ■ ■■■
01_data_ingestion.ipynb ■ ■■■ 02_data_cleaning.ipynb ■ ■■■ 03_eda_analysis.ipynb ■ ■■■
04_performance_analytics.ipynb ■ ■■■ 05_advanced_analytics.ipynb ■■■ scripts/ ■ ■■■
etl_pipeline.py ← Master ETL script ■ ■■■ live_nav_fetch.py ← mfapi.in fetcher ■ ■■■
compute_metrics.py ← Performance metrics ■ ■■■ recommender.py ← Fund recommender ■■■ sql/
■ ■■■ schema.sql ← CREATE TABLE statements ■ ■■■ queries.sql ← 10 analytical queries ■■■
dashboard/ ■ ■■■ bluestock_mf.pbix ← Power BI dashboard ■■■ reports/ ■ ■■■
Final_Report.pdf ■ ■■■ Presentation.pptx ■■■ README.md
```

7. Deliverables & Evaluation Rubric

#	Deliverable	Format	Weight	Evaluation Criteria
D1	ETL Pipeline Script	Python .py	**15%**	Code quality, handles errors, runs without manual steps
D2	SQLite Database	.db file	**10%**	Correct schema, all data loaded, queries run correctly
D3	EDA Notebook	.ipynb	**15%**	Depth of analysis, chart quality, insights documented
D4	Performance Metrics	.ipynb + CSV	**15%**	Mathematical accuracy, correct use of risk formulas
D5	Interactive Dashboard	.pbix or .twbx	**20%**	Design quality, functionality, all 4 pages complete
D6	Advanced Analytics	.ipynb	**10%**	VaR correctness, cohort analysis insight quality
D7	Final Report + Slides	.pdf + .pptx	**15%**	Professional quality, completeness, clarity of findings

7.1 Bonus Challenges (Optional +10 marks each)

- Deploy ETL pipeline as a scheduled script that auto-fetches NAV from mfapi.in every weekday at 8 PM
- Build a simple Streamlit web app as an alternative to Power BI for the dashboard
- Implement a Monte Carlo simulation to project NAV growth over 5 years with uncertainty bands
- Add a portfolio optimisation module (Markowitz Efficient Frontier) for 5 selected funds
- Create an automated email report generator that sends weekly performance summary as HTML

7.2 Common Mistakes to Avoid

Mistake	Correct Approach
■ Using simulated random NAV data without anchoring to real values	■ Use the real AMFI NAV anchor values provided in the dataset generator
■ Hard-coding file paths	■ Use pathlib.Path or os.path.join for cross-platform compatibility
■ Not handling weekends/holidays in NAV data	■ Use forward-fill (ffill) after reindexing to full date range
■ Computing CAGR with calendar days instead of trading days	■ Count actual rows (trading days) or use (252/n_days) annualisation
■ Dashboard with no filters/slicers	■ Every page must have at least 2 interactive slicers
■ Confusing AUM (lakh crore) with scheme-level AUM (crore)	■ Always include units in column names: aum_lakh_crore vs aum_crore
■ Uploading .db files >100MB to GitHub	■ Add *.db to .gitignore; upload schema.sql + instructions instead

8. Appendix: Dataset Schema Reference

01_fund_master.csv

Column	Type	Description
amfi_code	TEXT	AMFI unique scheme code (e.g. 125497 = HDFC Top 100 Direct)
fund_house	TEXT	AMC name (e.g. SBI Mutual Fund, HDFC Mutual Fund)
scheme_name	TEXT	Full official AMFI scheme name
category	TEXT	Equity / Debt / Hybrid
sub_category	TEXT	Large Cap / Mid Cap / Small Cap / Liquid / etc.
plan	TEXT	Regular or Direct
launch_date	DATE	Fund launch date (YYYY-MM-DD)
benchmark	TEXT	Official benchmark index
expense_ratio_pct	REAL	Annual expense ratio in % (e.g. 1.05)
exit_load_pct	REAL	Exit load % (0 for Liquid/Index funds)
fund_manager	TEXT	Name of primary fund manager
risk_category	TEXT	SEBI risk category: Low / Moderate / High / Very High
sebi_category_code	TEXT	Internal code: EC01=LargeCap, EC03=SmallCap, DC01=Liquid

02_nav_history.csv

Column	Type	Description
amfi_code	TEXT	Foreign key to fund_master
date	DATE	NAV date (business days only)
nav	REAL	NAV in Rs. (e.g. 892.4560). Anchored to real mfapi.in values.

04_monthly_sip_inflows.csv

Column	Type	Description
month	TEXT	YYYY-MM format
sip_inflow_crore	REAL	Total SIP inflows in Rs. crore (real AMFI data)
active_sip_accounts_crore	REAL	Number of actively contributing SIP accounts in crore
new_sip_accounts_lakh	REAL	New SIP registrations in that month (lakh accounts)
sip_aum_lakh_crore	REAL	Total SIP assets under management in Rs. lakh crore
yoy_growth_pct	REAL	YoY growth % in SIP inflows (computed)

07_scheme_performance.csv

Column	Type	Description

return_1yr_pct	REAL	1-year absolute return %
return_3yr_pct	REAL	3-year CAGR %
return_5yr_pct	REAL	5-year CAGR %
benchmark_3yr_pct	REAL	Benchmark index 3yr CAGR for comparison
alpha	REAL	Return above benchmark (return_3yr - benchmark_3yr)
beta	REAL	Sensitivity to market (1.0 = same as market)
sharpe_ratio	REAL	Risk-adjusted return (higher is better, >1 is good)
sortino_ratio	REAL	Like Sharpe but penalises only downside volatility
std_dev_ann_pct	REAL	Annualised standard deviation of daily returns (%)
max_drawdown_pct	REAL	Worst peak-to-trough decline (negative value)
morningstar_rating	INT	1-5 star rating (simulated, based on Sharpe)

08_investor_transactions.csv

Column	Type	Description
investor_id	TEXT	Unique investor identifier (INV000001 to INV005000)
transaction_date	DATE	Date of SIP/Lumpsum/Redemption
amfi_code	TEXT	Fund in which transaction occurred
transaction_type	TEXT	SIP / Lumpsum / Redemption
amount_inr	INT	Transaction amount in Indian Rupees
state	TEXT	Investor's state (12 Indian states covered)
city	TEXT	Investor's city
city_tier	TEXT	T30 (Top 30 cities) or B30 (Beyond Top 30) per AMFI classification
age_group	TEXT	18-25 / 26-35 / 36-45 / 46-55 / 56+
gender	TEXT	Male / Female
annual_income_lakh	REAL	Annual income in Rs. lakh
payment_mode	TEXT	UPI / Net Banking / Mandate / Cheque
kyc_status	TEXT	Verified (92%) / Pending (8%)

Note on Data Authenticity
All AMFI codes, fund names, fund houses, benchmarks, expense ratios, and AUM figures in these datasets are sourced from publicly available AMFI India data, mfapi.in, and financial news sources (Cafemutual, Business Standard). NAV values are anchored to real historical values and simulated forward using realistic return/volatility parameters consistent with actual market behaviour. Investor transaction data is synthetically generated but uses real geographic, demographic, and behavioural distributions observed in the Indian MF market. This project is intended for educational purposes and skill development at Bluestock Fintech.